

Elevate: A Decentralized, web3 enabled application with Smart contract

Blockchain Concepts and Technologies

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Academic Honesty Declaration

I declare the following to be true for this submission:

- I have completed the task during the designated time window and declare it to be exclusively my own work.
- I have not received or attempted to receive assistance in preparing this response from any other person during the assessment window.
- I have not provided, or offered to provide, assistance to any other student by any means during the assessment window.
- I have read and understand the National College of Ireland guidelines of Plagiarism.

Presentation link: [Blockchain project presentation-20230409_111454-Meeting Recording.mp4](#)

Live link: <https://elevatemarketingdapp.netlify.app/>

Introduction

Users can obtain marketing consultations using the decentralized Elevate application in a safe and transparent manner. Because the application was created utilizing Web3 technology, users' MetaMask wallets should be used for safe login. The application also establishes a connection to a deployed smart contract on the Sepolia test network Ethereum blockchain, allowing for safe and open Ethereum payment transactions. Users of the web-app get access to a range of marketing advisory services, such as marketing strategy, brand creation, and digital marketing tactics. With an easy-to-use UI, the program is also made to be user-friendly. Ultimately, by offering consumers safe and decentralized solutions, the Elevate decentralized application shows how Web3 technology and smart contracts have the potential to change a number of industries, including marketing and consultancy.

Decentralized apps, or dApps, operates autonomously with the use of smart contracts and run on peer-to-peer blockchain network. In the recent years Dapps have emerged as the most popular technology in the field of Blockchain Technology. The most well-known and often employed tools for creating dApps among them include Web3, MetaMask, Ethereum, Sepolia test network, smart contracts, Infura, and Truffle. The following tools has been used to develop this project:

- A) Web3:** Building decentralized apps that function on the blockchain is made possible by the Web3 suite of technologies and protocols. It offers a number of APIs and frameworks that enable programmers to communicate with the blockchain and create safe and open apps. Developers may use Web3 to design decentralized apps that are impervious to fraud, censorship, and manipulation.
- B) MetaMask:** With the help of the browser plugin MetaMask, users may utilize their Ethereum wallet to communicate with decentralized applications running on the Ethereum blockchain. It enables the sending and receiving of Ethereum and other ERC-20 tokens and offers secure login capabilities. Developers may test their apps on the Ethereum blockchain using MetaMask without releasing them to the public network (MetaMask, 2023).
- C) Ethereum:** Ethereum is a decentralized blockchain technology that supports smart contracts. It creates a peer-to-peer network that securely executes the smart contract (AWS, 2023) (Ethereum, 2023). It is a technology for developing apps, storing funds, and conducting transactions that are not governed by a central authority. It is also utilized as a cryptocurrency, with Ether serving as the Ethereum blockchain's native currency. Ethereum may be used by developers to create decentralized apps that are resistant to censorship, manipulation, and fraud.

- D) Sepolia test network:** The Sepolia test network is a safe and regulated environment where developers may test their decentralized apps before releasing them to the main Ethereum network. Before going live, this ensures that the program functions as expected and is free of defects or mistakes. Sepolia allows developers to test their applications in a safe and controlled environment without risking actual Ether (Alchemy, 2023).
- E) Smart contracts:** A smart contract is a straightforward program written in a programming language such as Solidity. It is deployed on the blockchain network and only executes when certain parameters are satisfied. It is used to automate the execution of agreements so that the concerned parties may obtain the results instantly without the involvement of any intermediaries, resulting in no time loss, a safe and transparent transaction (Investopedia, 2023).
- F) Infura:** Infura is a major blockchain IaaS platform that assists developers by providing immediate, scalable API access to Ethereum, allowing developers to link their decentralized apps to the Ethereum blockchain without hosting their own Ethereum node (Infura, 2023). It provides a dependable and scalable means for developers to link their apps to the blockchain, making it easier to construct and deploy decentralized applications. Developers may use Infura to connect to the Ethereum blockchain, send transactions, and interact with smart contracts without the requirement for a local Ethereum node.
- G) Truffle:** Truffle is an Ethereum blockchain development environment that allows developers to create decentralized apps by compiling and deploying smart contracts. Truffle contains a testing framework, a build pipeline, and a developer console in its smart contract development environment. It also supports major smart contract programming languages like Solidity. It offers a full range of resources for developing, testing, and deploying smart contracts, making it easy to develop powerful and creative dApps (GeeksforGeeks, 2023).

Setup and Configuration of tools:

This is a high-level description of how the above-mentioned technologies were set up and configured to develop a decentralized application on the Ethereum Sepolia test network:

Step 1) Install Node.js:

To install Node.js, we went to the Node.js website and downloaded the appropriate version for our operating system. And then we followed the website's installation instructions to install Node.js on our machine (Node.js, 2023).

Step 2) MetaMask:

To connect to our Ethereum-enabled decentralized application, we will utilize the MetaMask browser plugin. The web3 API will be injected into our JavaScript environment, allowing our Dapp to read from blockchain. We go to the MetaMask website and download the plugin to install MetaMask. After installed, create a new Ethereum wallet or import an existing one.

Step 3) Creating a web application:

We utilized ReactJs to construct our web application, and in order to create a new project in ReactJs, we must first open our terminal and enter the following command:

```
npx create-react-app elevate (Meta Open Source, 2023)
```

The command above will create a new ReactJs project in the elevate directory.

Step 4) Install Web3:

To add Web3 to our ReactJs project, we will use the following command in the terminal:

```
npm install web3 (npmjs, 2023)
```

This will include the installation of the Web3 library and its dependencies in our project.

Step 5) Inject Web3 into our application:

To inject Web3 into our React.js app, we'll construct a new Web3 object and connect it to an Ethereum node (Coinbase, 2023).

```
import Web3 from "web3/dist/web3.min.js";
const web3 = new Web3();
```

Step 6) Installing and setting up truffle:

To install Truffle, we will open our terminal and type the following command:

```
$ npm install -g truffle (GeeksforGeeks, 2023)
$ truffle unbox react (Truffle Suite, 2023)
```

This will create a framework to compile and deploy our smart contract on the blockchain network.

Step 7) Creating .env file and configuration:

We will create a .env file and inside that we will store the MetaMask recovery phrase and API key generated from the Infura.

Step 8) Write Smart contract:

To construct smart contracts, we will add a new elevate.sol file to our Truffle project's contracts directory. And In this file, write the Solidity code for our smart contract. The solidity file in this our smart contract contains of getting the enquiry or any suggestion from the customer or the purchase of services, and verification of their blockchain wallet making it secure and transparent.

Step 9) Compile and Migrate the Smart contract:

We will run the following command in our terminal to compile and deploy our smart contract to the Ethereum Sepolia test network.

```
truffle compile (Truffle Suite, 2023)
truffle migrate (Truffle Suite, 2023)
```

This will generate a elevate.json file which will contain the ABI of our contract. And we will place this file in our Elevate project to interact with the deployed smart contract.

Step 10) Interact with Smart contract:

To interact with smart contracts in our ReactJs application, we will build a new contract instance and invoke its functions with the use of Web3 object.

```
import Web3 from "web3/dist/web3.min.js";
import Elevate from "./Elevate.json";
```

```
const web3 = new Web3();
const contractaddress = new web3.eth.Contract( Elevate.abi, Elevate.networks[networkId].address);
```

Interaction between all of the components utilized in this project:

The basic architecture of the given system includes numerous components that work together to enable interaction with a smart contract on the Sepolia test network. Users should connect their MetaMask wallet to the Elevate Marketing application and utilize the smart contract to make network transactions.

The architecture is made up of the following components:

- **Smart Contract:** This is the most significant component, as it holds the business logic and data for the application. We create the smart contract in Solidity and build it with the Truffle framework before deploying it to the Sepolia test network, with which our decentralized application may communicate.
- **Sepolia Test Network:** This is the blockchain network where our smart contract is implemented and tested. The network is used to mimic real-world transactions and interactions with the smart contract.

- **Truffle:** It is a development platform that provides developers with tools for building and deploying smart contracts. We utilized Truffle to compile and deploy the smart contract to the Sepolia test network in this case. When a smart contract is deployed, a json file is created that includes the ABI and functions of the smart contract and can be read by our application to communicate with the smart contract.
- **Elevate Marketing Application:** This is the user-facing application that allows interaction with the Sepolia test network's smart contract. Users may use the app to link their MetaMask wallet and make network transactions.
- **MetaMask Wallet:** This is a digital wallet that allows users to store, manage, and connect with decentralized apps. The Elevate Marketing application authorizes and sends transactions to the Sepolia test network using MetaMask.

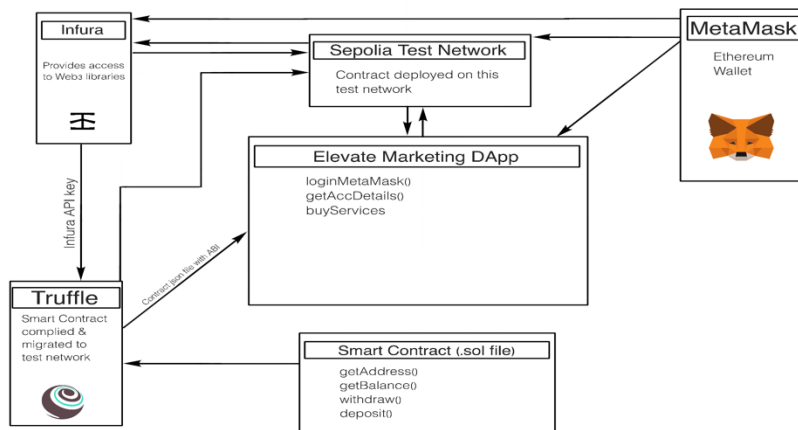
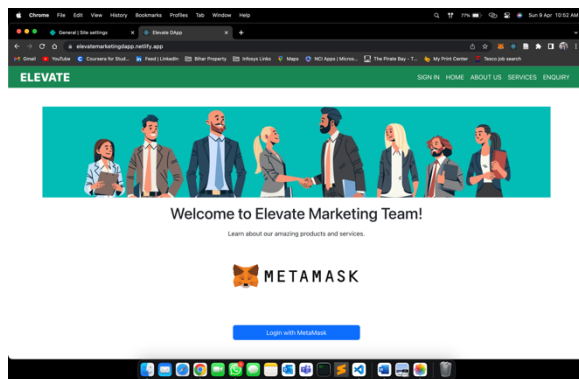
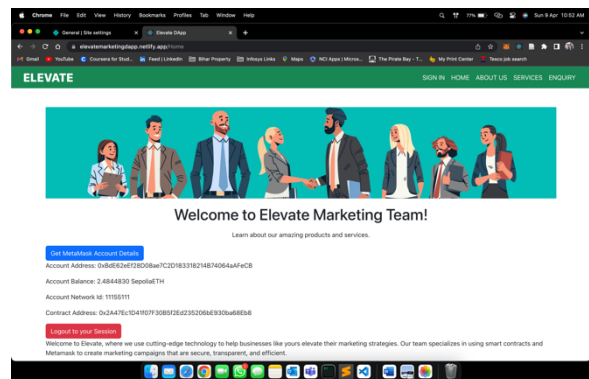


Fig 1. Architectural Diagram

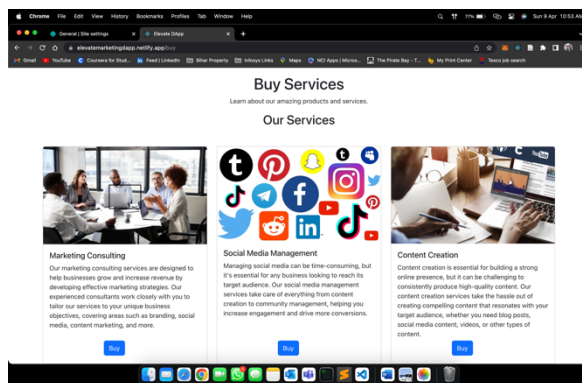
Screen Shots of the application:



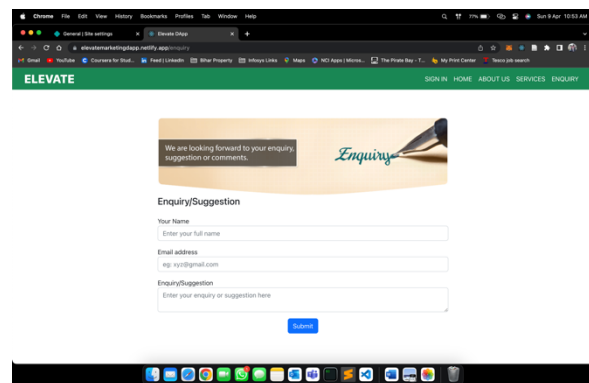
Screenshot. 1



Screenshot. 2



Screenshot. 3



Screenshot. 4

- Screenshot 1 shows the functionality of login through MetaMask Wallet which allows user to access the services of the application.
- Screenshot 2 shows the functionality of getting MetaMask account details and getting the Smart Contract address.
- Screenshot 3 shows the functionality of purchasing the services provide by the decentralized application, upon clicking on the buy button it will deduct some Ethereum from the wallet.
- Screenshot 4 shows the functionality of getting the enquiry or any suggestion from the customer.

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Use and Purpose of Smart Contracts

Smart contracts automate the process of verifying, carrying out, and upholding contractual agreements between parties. They are self-executing digital contracts. Without the need of intermediaries, they may conduct safe, traceable, and tamper-proof transactions due to blockchain technology.

Smart contracts' main goal is to automate the implementation of contractual agreements, eliminating the need for human interaction and lowering the risk of mistakes or fraudulent activity. Many industries, including banking, supply chain management, real estate, and healthcare, to mention a few, may employ smart contracts.

One of the most significant impacts of smart contracts is the ability to create tokens, NFTs, and smart contract-enabled applications. Tokens can represent anything of value, such as currencies, commodities, or even loyalty points, and can be traded on decentralized exchanges. NFTs, or non-fungible tokens, allow for the creation and sale of unique digital assets, such as art or collectibles, on blockchain platforms. Smart contract-enabled applications, or decentralized applications (dApps), allow for the development of various applications that run on blockchain networks, providing a new level of security, transparency, and decentralization.

Yet, there are also social and ethical issues with using smart contracts. One issue is the potential for smart contracts to maintain or worsen already-existing inequities, particularly in areas like land ownership or access to financial services. As they offer an anonymous and decentralized platform, smart contracts raise worries about their potential to facilitate unlawful actions like financial fraud or terrorism funding.

Two examples of state-of-the-art applications of smart contracts are:

Chainlink: Chainlink is a decentralized oracle network that links smart contracts to data and events from the real world. It makes it simpler to create dApps that use external data by enabling safe access to off-chain data for smart contracts. Chainlink doesn't run on its own blockchain or inside of another blockchain because it is blockchain agnostic. Instead, it concurrently operates on many blockchains and offers incentives to its users to include and utilize Chainlink data in their blockchain smart contracts. The oracle network of Chainlink may be utilized to predict anything from the forecast to the outcomes of the presidential election. The current market valuation of Chainlink is \$7.23 USD, with the 24 hour market cap of \$146,292,156 USD.

MakerDAO: MakerDAO is a decentralized lending platform that enables users to utilize their cryptocurrency holdings as leverage to create Dai, a stable coin tied to the US dollar. As the platform is totally controlled by smart contracts, the loan process is transparent and secure. The borrower is required to pay interest on the loans, but if the value of the crypto collateral drops too low, it will be sold to cover the debt. The Maker Protocol has undergone the most significant modification due to the fact that it now takes any Ethereum-based collateral in order to produce DAI, provided that it has been accepted by MKR holders and has been granted specified, equivalent Risk Parameters through the Maker decentralized governance process. According to recent reports, DAI instability may be the end of Ethereum's favorite cryptocurrency.

Finally, by automating the execution of contracts, producing tokens and NFTs, and enabling the creation of decentralized apps, smart contracts have the potential to transform a number of sectors. Yet there are also social and ethical issues that must be resolved, such possible injustices and unlawful acts. Modern apps like Chainlink and MakerDAO are examples of how smart contracts have the ability to disrupt established institutions and open up new possibilities.